



Theoretical study of Ti:LiNbO₃ strip waveguides in the mid-infrared

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Abstract. The transverse magnetic and transverse electric mode size and effective refractive index as functions of the Ti-strip initial width W , diffusion temperature T and Ti-strip initial thickness H in c -cut Ti-diffused LiNbO₃ waveguides have been theoretically studied at wavelengths of 3.0 and 3.39 μm on the basis of modified expressions for the waveguide surface index increment. The single-mode, multiple-mode and cut-off conditions have been identified from those effective refractive index plots. These mid-infrared numerical results are discussed comparatively with near-infrared data reported earlier.

1. Introduction

The interest in the Ti:LiNbO₃ waveguide in integrated optics no longer needs to be justified. For its application in many waveguide devices, it is essential to know the mode distribution, effective refractive index, single-mode and cut-off conditions at a given wavelength. To date, much theoretical and experimental work on these aspects has been carried out at a wavelength of around 1.55 μm . In recent years the Ti:LiNbO₃ waveguide applied in the mid-infrared (MIR) region has also received much attention because of its application as a radiation source in chalcogenide and fluoride fibre testing [1], high-resolution molecule analysis [2] and sensing of many atmospheric trace gases such as HF, H₂CO₃, CH₄, NO, N₂O, SO₂ and CO [3]. In the last decade, some experimental data about the fabrication of a single-mode waveguide in the MIR region have been reported in several papers [4–7]. However, the corresponding theoretical work, suitable for the MIR region, on the dependences of the mode size, effective refractive index and single-mode condition on the waveguide fabrication parameters has not been presented so far, to our best knowledge. In this paper, the guided mode and effective refractive index in Ti:LiNbO₃ waveguides at 3.0 and 3.39 μm were analysed by the variational method after the expressions for the waveguide surface index increment had been modified. The mode sizes and effective refractive indices as functions of the controllable fabrication parameters such as initial Ti-strip width, diffusion temperature and initial Ti-strip thickness were

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numerically calculated for two lower-order modes, allowing us to identify the cut-off condition, single-mode condition and multiple-mode condition.

2. Numerical method and discussion

2.1. The modification of expressions for the increase in waveguide surface index

The calculation procedure for analysing the Ti:LiNbO₃ waveguide with the variational method has been described in detail in an earlier paper [8]. The variational equations for the propagation constant for a *c*-cut Ti-diffused LiNbO₃ waveguide are as follows: for the transverse electric (TE_{*n*0}) mode,

$$\beta^2 = k_0^2 n_{\text{ob}}^2 + \frac{2^{3/2}(n_{\text{os}}^2 - n_{\text{ob}}^2)k_0^2}{W_x} \left(1 + \frac{W_y^2}{2d_{\text{yo}}^2}\right)^{-3/2} Q - \frac{2n+1}{W_x^2} - \frac{3}{W_y^2}, \quad (1)$$

and, for the transverse magnetic (TM_{*n*0}) mode,

$$\beta^2 = k_0^2 n_{\text{eb}}^2 + \frac{2^{3/2}(n_{\text{es}}^2 - n_{\text{eb}}^2)k_0^2}{W_x} \left(1 + \frac{W_y^2}{2d_{\text{ye}}^2}\right)^{-3/2} Q - \frac{2n+1}{W_x^2} - \left(\frac{n_{\text{eb}}}{n_{\text{ob}}}\right)^2 \frac{3}{W_y^2}, \quad (2)$$

where $Q = \int_0^{x_z} g(2x/W) \psi_n^2(2^{1/2}x/W_x) dx$, in which the upper limit x_z of the integral satisfies the relation $g(2x/W) = 0$; the expressions for $g(2x/W)$ and ψ_n can be found in [9]. In the preceding two equations, x and y are the coordinates along the width and depth directions respectively of the waveguide, k_0 is the wavenumber in free space, n_{eb} and n_{ob} are the substrate indices, n_{es} and n_{os} are the surface indices, d_{ye} and d_{yo} are the diffusion depths of the waveguide, and the subscripts e and o represent extraordinary and ordinary rays respectively.

The above two equations involve the question of extremum value solution for a binary function. The optimum values of mode sizes W_x and W_y correspond to the largest value of β^2 , namely β_m^2 , the corresponding effective refractive index can then be determined by $N_{\text{eff}} = (\lambda/2\pi)\beta_m$.

The key question of solving equations (1) and (2) is the determination of the waveguide surface index n_{es} or n_{os} , that is how should the waveguide surface index increment Δn_{es} or Δn_{os} be fixed. The original expressions for Δn_{es} or Δn_{os} , which are only suitable for the near-infrared (NIR) range (0.6–1.6 μm), cannot be applied to the MIR area; so they need modifying to some extent. According to Fouchet *et al.* [10], the equations for calculating Δn_{es} and Δn_{os} suitable for the NIR region are expressed as

$$\begin{aligned} \Delta n_{\text{es}} &= \left(B_0(\lambda) + B_1(\lambda) \frac{H}{d_{\text{ye}}}\right) \left(\frac{H}{d_{\text{ye}}}\right)^{\alpha_e}, \\ \Delta n_{\text{os}} &= \left(B_0(\lambda) + B_1(\lambda) \frac{H}{d_{\text{yo}}}\right) \left(\frac{H}{d_{\text{yo}}}\right)^{\alpha_o}, \end{aligned} \quad (3)$$

where the fitted constants $\alpha_e = 0.83$ and $\alpha_o = 0.53$; for the ordinary ray,

$$\begin{aligned} B_0(\lambda) &= 0.0653 - 0.0315\lambda + 0.00709\lambda^2, \\ B_1(\lambda) &= 0.478 + 0.464\lambda - 0.348\lambda^2, \end{aligned} \quad (4)$$

and, for the extraordinary ray,

$$\begin{aligned} B_0(\lambda) &= 0.385 - 0.43\lambda + 0.171\lambda^2, \\ B_1(\lambda) &= 9.13 + 3.85\lambda - 2.49\lambda^2. \end{aligned} \quad (5)$$

For the MIR region, the expressions for Δn_{es} and Δn_{os} can be modified by expanding the above second-order polynomials $B_0(\lambda)$ and $B_1(\lambda)$ to third order. For the ordinary ray,

$$\begin{aligned} B_0(\lambda) &= 0.0653 - 0.0315\lambda + 0.00709\lambda^2 + a_3\lambda^3, \\ B_1(\lambda) &= 0.478 + 0.464\lambda - 0.348\lambda^2 + b_3\lambda^3, \end{aligned} \quad (6)$$

and, for the extraordinary ray,

$$\begin{aligned} B_0(\lambda) &= 0.385 - 0.43\lambda + 0.171\lambda^2 + c_3\lambda^3, \\ B_1(\lambda) &= 9.13 + 3.85\lambda - 2.49\lambda^2 + d_3\lambda^3, \end{aligned} \quad (7)$$

in which a_3 , b_3 , c_3 and d_3 are the coefficients awaiting solution. It is obvious that these unknown coefficients will be fixed as soon as any two values of Δn_{es} or Δn_{os} are known. These two Δn_{es} or Δn_{os} should correspond to two waveguides with different diffusion temperatures or initial Ti-strip thicknesses. Otherwise, if the same diffusion temperatures and initial Ti-strip thicknesses are considered, no solution for (a_3, b_3) or (c_3, d_3) will be found in equation (6) or (7) because the two waveguides will have same ratios of H/d_{ye} and H/d_{yo} even though the considered wavelengths are different. Experimental fabrication parameters of some individual single-mode and multiple-mode waveguides at 3.0 and 3.39 μm reported earlier [4–7] make it possible to find two individual Δn_{es} or Δn_{os} values. The fixing of two individual Δn_{es} or two individual Δn_{os} was accomplished by calculating the relations for the effective refractive index of two lower-order modes against the Ti-strip width according to equation (1) or (2). Every calculation result allows us to identify one single-mode region of the Ti-strip width. Subsequently, a comparison between the theoretically resulting single-mode range of the Ti-strip width and the experimental data was performed until satisfactory agreement was reached. Finally, the values of the coefficients a_3 , b_3 , c_3 and d_3 can be solved by inserting the relevant λ and corresponding Δn_{es} and Δn_{os} into equations (3), (6) and (7).

2.2. Results and discussion

The experimental data employed to modify the above expressions of waveguide surface increment are collected in table 1. In the modification procedure, two sets of experimental data $\lambda = 3.39 \mu\text{m}$, $T = 1060^\circ\text{C}$, $H = 140 \text{ nm}$ and $t = 31 \text{ h}$ [4] and $\lambda = 3.39 \mu\text{m}$, $T = 1050^\circ\text{C}$, $H = 160 \text{ nm}$ and $t = 35 \text{ h}$ [5] were used for the modification of Δn_{os} , while the data sets $\lambda = 3.39 \mu\text{m}$, $T = 1050^\circ\text{C}$, $H = 160 \text{ nm}$ and $t = 35 \text{ h}$ [5] and $\lambda = 3.0 \mu\text{m}$, $T = 1060^\circ\text{C}$, $H = 160 \text{ nm}$ and $t = 31 \text{ h}$ [6, 7] were employed to modify Δn_{es} . Accordingly, the modified $B_0(\lambda)$ and $B_1(\lambda)$ are, for the ordinary ray,

$$\begin{aligned} B_0(\lambda) &= 0.0653 - 0.0315\lambda + 0.00709\lambda^2 + 0.00271\lambda^3, \\ B_1(\lambda) &= 0.478 + 0.464\lambda - 0.348\lambda^2 - 0.152\lambda^3, \end{aligned} \quad (8)$$

Table 1. Calculated single-mode, cut-off and multiple-mode conditions at wavelengths of 3.0 and 3.39 μm for *c*-cut Ti:LiNbO₃ waveguides. Also listed are the experimental data taken from [4–7].

	Single-mode condition	Cut-off condition	Multiple-mode condition	Wavelength (μm)	Mode	Other diffusion parameters
Calculated result for W (μm)	11.4–31.2	< 11.4	> 31.2	3.39	TE	$T = 1060^\circ\text{C}$, $H = 140\text{ nm}$, $t = 31\text{ h}$
Experimental data [4, 5] for W (μm)	13.1–34.7	< 13.1	> 34.7	3.0	TE	$T = 1050^\circ\text{C}$, $H = 160\text{ nm}$, $t = 35\text{ h}$
	20, 30	< 10		3.39	TE	$T = 1060^\circ\text{C}$, $H = 140\text{ nm}$, $t = 31\text{ h}$
	18			3.39	TE	$T = 1050^\circ\text{C}$, $H = 160\text{ nm}$, $t = 35\text{ h}$
	11.9–30.2	< 11.9	> 30.2	3.39	TM	$T = 1050^\circ\text{C}$, $H = 160\text{ nm}$, $t = 35\text{ h}$
Calculated result for W (μm)	10.6–27.5	< 10.6	> 27.5	3.0	TM	$T = 1060^\circ\text{C}$, $H = 160\text{ nm}$, $t = 31\text{ h}$
Experimental data [5–7] for W (μm)	18			3.39	TM	$T = 1050^\circ\text{C}$, $H = 160\text{ nm}$, $t = 35\text{ h}$
	15, 17.5, 20, 22.5			3.0	TM	$T = 1060^\circ\text{C}$, $H = 160\text{ nm}$, $t = 31\text{ h}$
	1038–1080	< 1038	> 1080	3.0	TE	$W = 25\text{ }\mu\text{m}$, $H = 160\text{ nm}$, $t = 35\text{ h}$
	1042–1073	< 1042	> 1073	3.0	TM	$W = 18\text{ }\mu\text{m}$, $H = 160\text{ nm}$, $t = 31\text{ h}$
Caculated result for T ($^\circ\text{C}$)	< 220	> 220		3.0	TE	$W = 25\text{ }\mu\text{m}$, $T = 1050^\circ\text{C}$, $t = 35\text{ h}$
	145–180	< 145	> 180	3.0	TM	$W = 18\text{ }\mu\text{m}$, $T = 1060^\circ\text{C}$, $t = 31\text{ h}$

and, for the extraordinary ray,

$$\begin{aligned} B_0(\lambda) &= 0.385 - 0.43\lambda + 0.171\lambda^2 - 0.041\lambda^3, \\ B_1(\lambda) &= 9.13 + 3.85\lambda - 2.49\lambda^2 + 1.713\lambda^3. \end{aligned} \quad (9)$$

Under these modifications and for the above-mentioned experimental parameters, the calculated dependences of effective refractive index and corresponding mode size on the Ti-strip width W are illustrated in figures 1 and 2 respectively. From figure 1 the single-mode region for W can be readily analysed on the basis of $n_{\text{eb,ob}} < N_{\text{eff}} < n_{\text{es,os}}$, in which n_{eb} and n_{ob} are the substrate indices. For a wavelength of $3.0\ \mu\text{m}$, $n_{\text{ob}} = 2.161\ 470$ and $n_{\text{eb}} = 2.095\ 659$; for wavelength of $3.39\ \mu\text{m}$, $n_{\text{ob}} = 2.144\ 726$ and $n_{\text{eb}} = 2.081\ 577$ [11]. In figure 1(a), it is obvious that a waveguide with Ti strip $13.1\text{--}34.7\ \mu\text{m}$ wide supports only the fundamental mode. When the stripe width W is increased to greater than $34.7\ \mu\text{m}$ the multiple-mode region is approached. The single-mode regions of the other three figures are $11.4\text{--}31.2$, $10.6\text{--}27.5$ and $11.9\text{--}30.2\ \mu\text{m}$ for figures 1(b), (c) and (d) respectively. These theoretical results are explicitly listed in table 1. One can see that the theoretical TE and TM single-mode condition for W easily covers the experimental data, indicating that the theoretical results and experimental data are self-consistent.

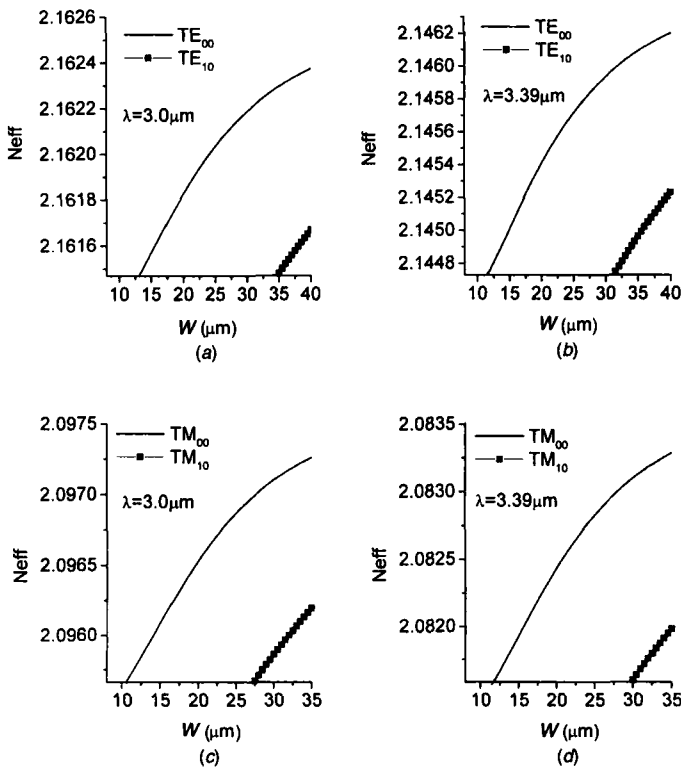


Figure 1. Effective refractive indexes versus initial Ti strip width W : (a) TE mode at $3.0\ \mu\text{m}$, with the fabrication parameters $T = 1050^\circ\text{C}$, $H = 160\ \text{nm}$ and $t = 35\ \text{h}$; (b) TE mode at $3.39\ \mu\text{m}$ with $T = 1060^\circ\text{C}$, $H = 140\ \text{nm}$ and $t = 31\ \text{h}$; (c) TM mode at $3.0\ \mu\text{m}$, with $T = 1060^\circ\text{C}$, $H = 160\ \text{nm}$ and $t = 31\ \text{h}$; (d) TM mode at $3.39\ \mu\text{m}$ with $T = 1050^\circ\text{C}$, $H = 160\ \text{nm}$ and $t = 35\ \text{h}$.

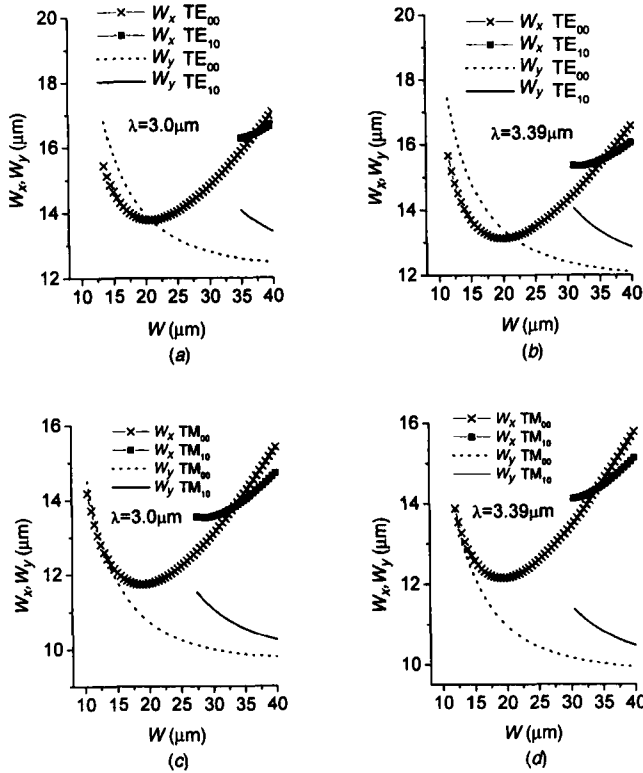


Figure 2. Mode size, corresponding to the effective refractive indices illustrated in figure 1, versus the initial Ti-strip width W .

The variation in mode size with W , as shown in figure 2, reveals a similar behaviour to that in the NIR region (1563 nm) [8]. Nevertheless, considerable differences between the MIR and NIR still exist. For example, the mode size of the MIR is obvious larger than that of the NIR owing to the longer wavelength considered, the larger strip width and the smaller surface index increment (and hence weaker confinement) to the guided mode. The smaller surface index increment of the MIR waveguide also results in a looser single-mode condition than for the NIR waveguide. It can be seen from table 1 that the span covered by the single-mode region of W exceeds 15 μm for the MIR TM mode and 20 μm for the MIR TE mode, while the data for NIR (1563 nm) TM and TE single-mode waveguides are only several microns and ten microns respectively [8].

Based on the parameters $W = 25 \mu\text{m}$, $T = 1050^\circ\text{C}$, $H = 160 \text{ nm}$ and $t = 35 \text{ h}$ and the parameters $W = 18 \mu\text{m}$, $T = 1060^\circ\text{C}$, $H = 160 \text{ nm}$ and $t = 31 \text{ h}$, the dependences of the diffusion temperature and initial Ti-strip thickness on the effective refractive index and mode size of two lower-order TE and TM modes were also calculated for a wavelength of 3.0 μm . The calculated results are shown in figures 3 and 4. The related data on single-mode, cut-off and multiple-mode conditions are also listed in table 1. Under the considered parameters, the diffusion temperature range 1038–1080 $^\circ\text{C}$ is the single-mode condition for the TE mode and 1042–1073 $^\circ\text{C}$ is the single-mode condition for the TM mode. For the thickness dependence, as soon as the initial Ti-strip thickness is thinner than 220 nm, the

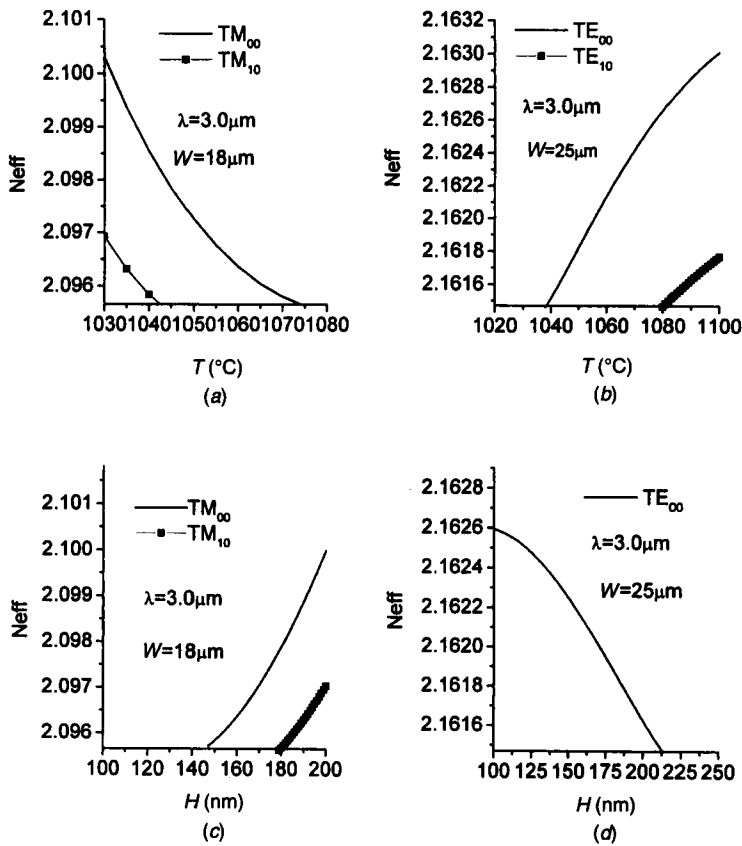


Figure 3. (a), (b) Effective refractive index versus diffusion temperature T at a wavelength of $3.0 \mu\text{m}$ (a) for the TM mode and (b) the TE mode with the parameters (a) $W = 18 \mu\text{m}$, $H = 160 \text{ nm}$ and $t = 31 \text{ h}$ and (b) $W = 25 \mu\text{m}$, $H = 160 \text{ nm}$ and $t = 35 \text{ h}$; (c), (d) effective refractive index versus initial Ti-strip thickness H at a wavelength of $3.0 \mu\text{m}$ (c) for the TM mode and (d) the TE mode with the parameters (c) $W = 18 \mu\text{m}$, $T = 1060^{\circ}\text{C}$ and $t = 31 \text{ h}$ and (d) $W = 25 \mu\text{m}$, $T = 1050^{\circ}\text{C}$ and $t = 35 \text{ h}$.

waveguide will support only the fundamental TE mode; when it is thicker than 220 nm , the waveguide will be cut off. For the TM mode, a waveguide with a thickness of $145\text{--}180 \text{ nm}$ supports only the TM_{00} mode; below 145 nm the waveguide does not guide; above 180 nm it will support more TM modes.

A rough comparison between the diffusion temperature and Ti-strip thickness dependences on the effective index for the MIR and NIR (1563 nm for example) allow us to find that all the various tendencies of the TE and TM effective indices of the MIR waveguide are the same as those of the 1563 nm waveguide [8] except for the thickness dependence for the TE mode shown in figure 3(a), where the effective index decreases instead of increases as reported in [8], with increase in the Ti-strip thickness. A difference between the mode size behaviours for the MIR and NIR (at 1563 nm) is observed for TE polarization (figures 4(b) and (d)). In figure 4(b), the mode size decreases instead of increases as in the case at 1563 nm [8], with increase in the diffusion temperature; in figure 4(d), the mode size increases instead of decreases with increase in the thickness H .

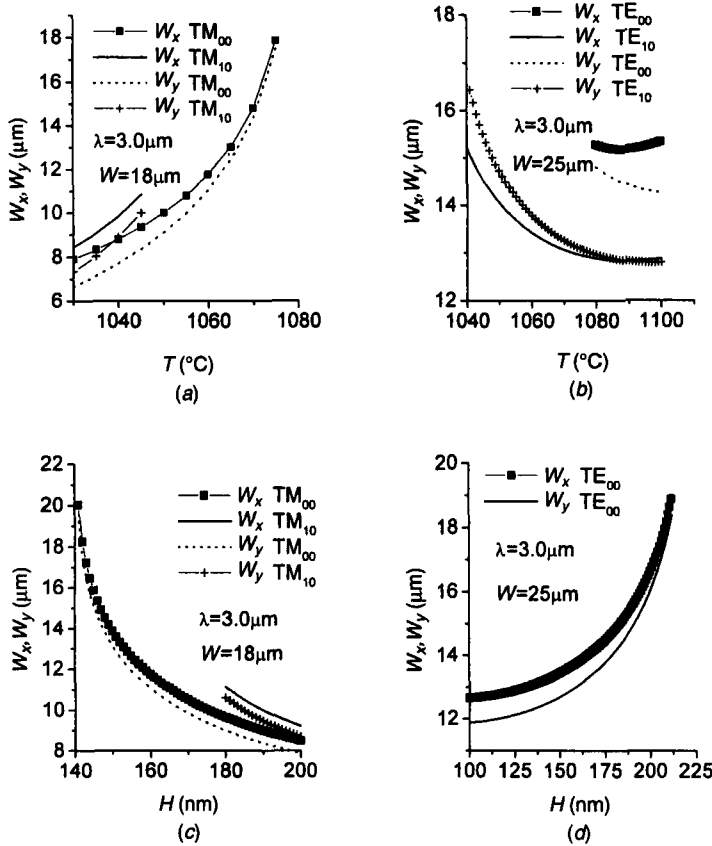


Figure 4. Mode size, corresponding to the effective refractive indices illustrated in figure 3, versus diffusion temperature T and initial Ti-strip thickness H .

3. Conclusions

We have demonstrated the numerical results of c -cut Ti: LiNbO₃ waveguides for application in the MIR region. The expressions for the waveguide surface index increment have been modified on the basis of experimental data published earlier. The dependences of mode sizes and corresponding effective refractive indices on controllable fabrication parameters at 3.0 and 3.39 μm wavelengths have been calculated for two lower-order TM and TE modes, allowing us to identify single-mode conditions, which are self-consistent with the experimental data. The present numerical results may offer to some extent guidance to the fabrication of a practical MIR single-mode Ti: LiNbO₃ waveguide. The dependences of the controllable waveguide fabrication parameters on the mode size and effective refractive index of MIR waveguides reveal some differences from those of NIR waveguides.

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